GROUP I

Final Project Report

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PROJECT NAME: KITCHENSYNC

PROJECT SPONSOR: TIMOTHY MACIAG

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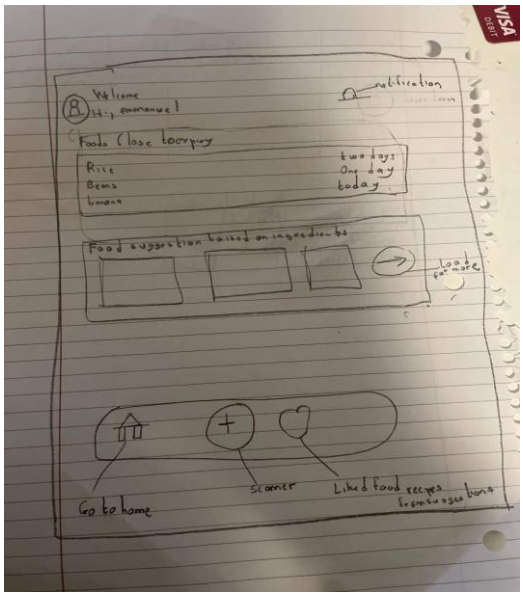
INTRODUCTION

During our project planning, we explored the idea of creating a web app that keeps record of a user's food inventory, separating it into three categories, a pantry, a fridge and a freezer, after this the web app keeps track of the inventory, providing expiration dates for every product and providing recipe suggestions based off the provided inventory. With this innovation we were looking to the combat the issue of household food management, the disorganization that comes with prepping for a meal and how a lot of food items are forgotten and left to go bad, we tried the bridge a gap between a user and their inventory, leading to a reduction in food wastage and saving money by better interpretation and management of household food inventory.

The idea was to bridge the gap between proper food purchasing and efficient consumption by building a system that cut down the wastage, saved money, planned and recommended meals based off food inventory. We as a group really liked the idea and embraced the challenge of creating this web app. Our team is an inexperienced one, made up of beginner coders, especially for the architecture and skillset needed for a project like this, facing constraints like time, as we had a timeframe of two months and four different packaged schedules for each of us to work around, nevertheless we got to work.

A main component of our initial design thought was the implementation of scanners, to scan the barcode of a product to add it into the inventory, which would auto-generate and input the expiration date of the product. A home page icon was also added at the bottom center of every page, to offer user flexibility. Our initial design was meant to have just one “plus” function to add all the items but later we pivoted to adding different “plus” functions for three different storages, a fridge, a freezer and a pantry whilst also allowing the user to add a new storage space if the wanted to.

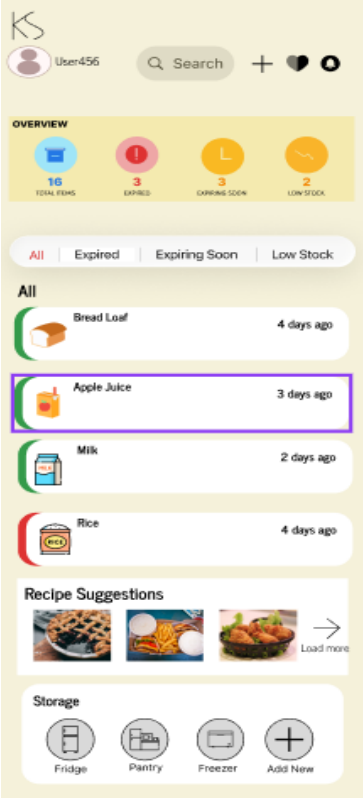
Firstly, a user enters the food item into the app using a “plus (+)” function. To address the issue of wastage, the inventory was separated into categories is to provide the user with options and affording flexibility instead of dumping a whole shopping cart into one spot. These categories are the fridge, freezer, and pantry. For the expiration tracking functionality, a dashboard displays the expiration on the homepage, the expiration function is separated into three, the upcoming expiration dates (expiring soon), the products not yet expiring but low on (low stock) and the already past expiration dates (expired). A search function is also implemented, to search the inventory for a specific product; a “favorites” page was also added, allowing the user to like a recommended recipe and come back to it in the future. For the recommendation function we planned to implement APIs, a functionality that allows us to connect to a database that contains information about different meals and the ingredients needed to make them.



(Lo-fi diagram of original dashboard design)

After the first scrum report, we received feedback on the implementation of a scanner feature, suggesting that it was out of scope for the size of the project, and that any

implementation of AI for recipe auto-generation wasn't advisable. With the second scrum report, we had a hi-fi prototype of the web application ready. The feedback was largely positive. Users felt the icons were cluttered at the top of the dashboard, and the app logo also needed some work. There was positive feedback on the design; the color scheme, iconography, text color, and size were good. By our next iteration, we decluttered the dashboard, resized and restructured the expiration section of the dashboard, improved the search functionality, by adding a search function for each storage section, and changed the app logo. The scanning function was abandoned which had a ripple effect on the auto-generated expiration dates; users would have to input the expiration date.

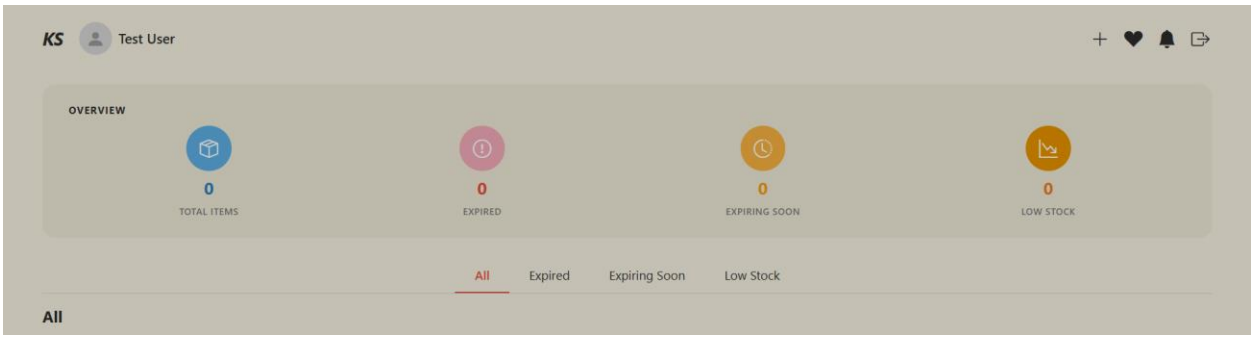


(High-fidelity prototype of the dashboard designed on Figma)

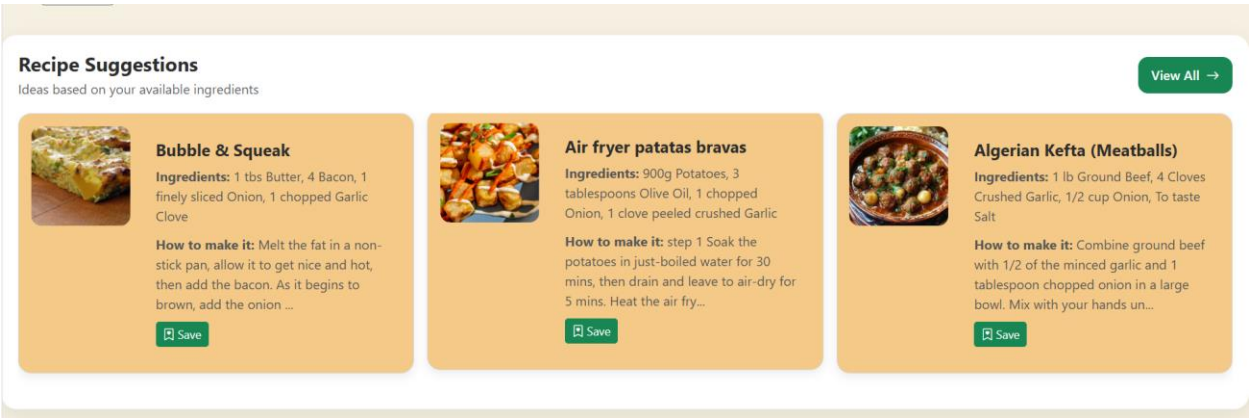
By the third scrum report, we had the entire front-end coded and ready, work on the back-end hadn't really begun, feedback wasn't really encouraging as there was no functionality

implemented, we were advised against implementing the recipe recommendation functionality, rather abandon it entirely and focus on delivering an MVP without this function as it would still meet the requirements to be considered an MVP. We conveyed as a group and deliberated on want we wanted to do as a team. We decided to go ahead with the implementation of the API, as it was one of the main functionalities which made our web app distinctive and tackled a big question mark in our problem definition. The API/database implemented is known as MealDB, which was recommended by our scrum master. Providing our web app with recipe recommendations and images to go with as well as images for every single item entered in the inventory.

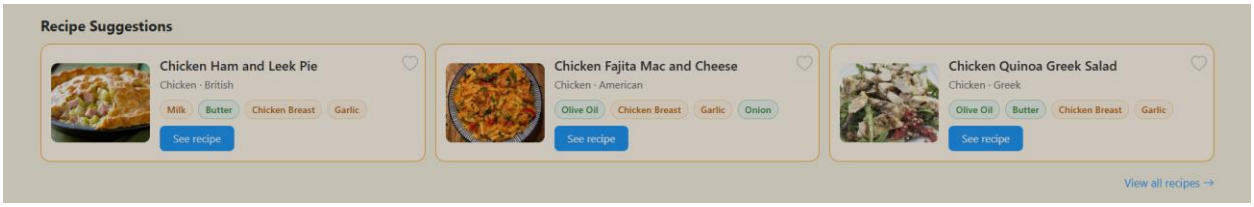
The outcome was great; we achieved a working MVP that tackled all the problems we sought to solve. A web app that allows a signed in user to input all their food items, keep track of it, get in app notifications on expirations and low stock items, get recipe recommendations based on the inventory focused on the items soon to expire, so as to prevent it from going bad. With this, we believe users will be able to save more money using this app, by cutting down unnecessary grocery expenditures. After ditching the scanner idea, we stuck with the idea of the separation of the inventory into three parts, a fridge, a freezer and a pantry whilst adding an additional “plus” function to create a new type of storage if the so please, a search function was added to every single storage section and a “favorites” page too was created too, where every liked recipe is saved. A page dedicated to the recipe suggestions was added. This page is accessible through a “view all recipes” button on the dashboard.



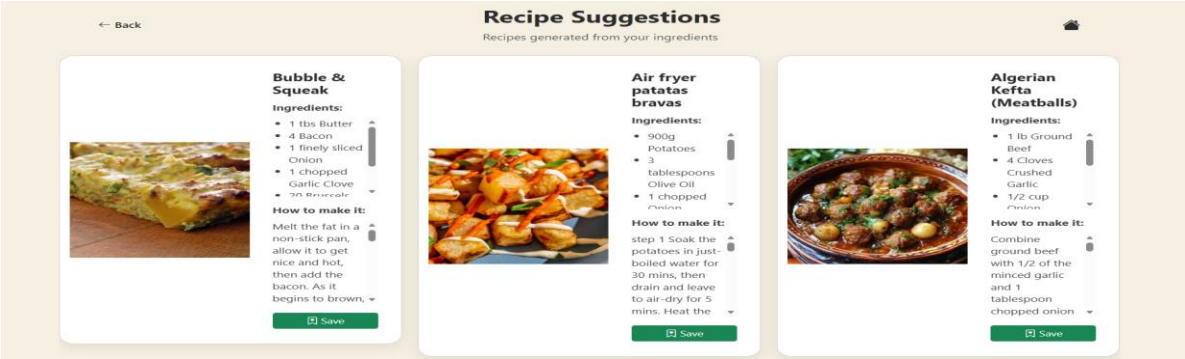
(view of Top Bar and “Overview” section on dashboard page)



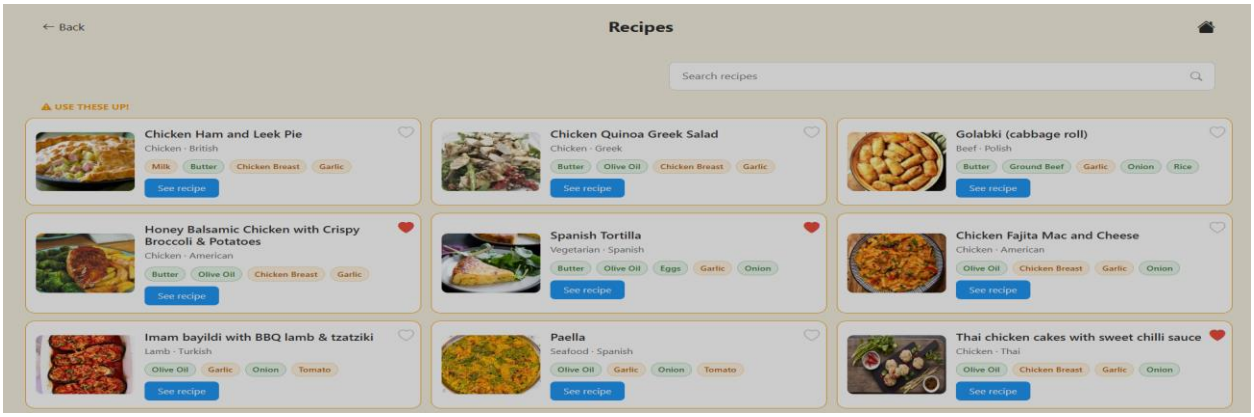
(view of “Recipe Suggestions” on dashboard page from demo/initial state)



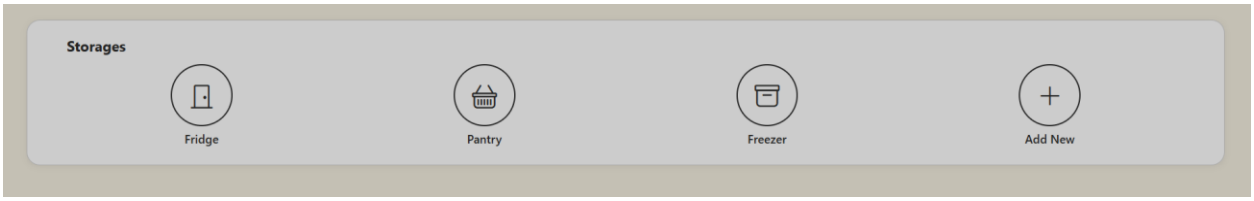
(view of “Recipe Suggestions” on dashboard after UI changes/current state)



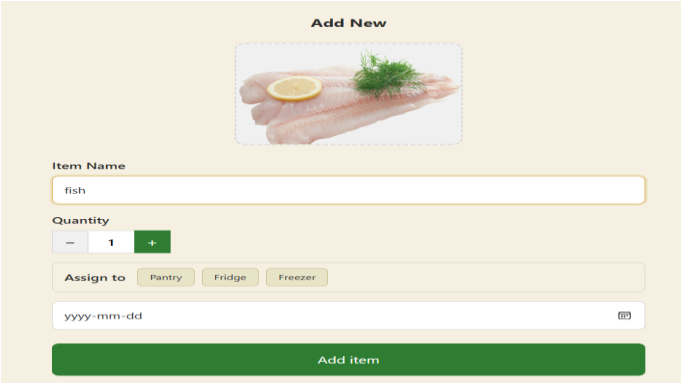
(view of “All Recipe Suggestions” page from demo/initial state)



(view of “All Recipe Suggestions” page after UI changes/current state)



(view of “Storages” section on dashboard page)



(view of “Add Item” page)

Reflecting on the delivered MVP, we are proud of the work we achieved in this short timeframe. We learned so much, having to work around our schedules and debate on ideas and decisions, reflecting feedback, creating and editing, whilst maintaining focus on our main goal and the task we set out to achieve. Working in a group can be difficult; you learn quickly that everyone has different ideas and different ways of doing things to achieve the same goal, so you quickly learn that communication is central to it all. Communication played a big role in the success of our project, effectively communicating, constant in-person meetings and virtual meetings when necessary and convenient really sped up and raised the quality of the process of sharing ideas and opinions. Sharing a centralized workspace was integral, utilizing GitHub we were able to keep up with what everyone was doing, using a kanban board to track the progression of our project tasks. This helped us know when to meet up, discuss, plan, and decide on important tasks to ensure successful scrum reports at every juncture. As the project progressed, team experienced some frustration, schedules became more difficult to work around due to individual workloads, our knowledge base was also limited, none of us were advanced coders, so the scope of whatever we could implement with code was limited. This reflected our

final MVP delivered, leaving whatever ideas that had to be scrapped for future implementation. Reflecting on the current state of the MVP, there's some good ideas on scaling it up for an MVP 2, the implementation of the scanner would really be ideal, a general search bar, that searches every page of the web app and the addition of an external notification system.

OUR TEAM'S OVERALL SELF REFLECTION

During our project planning we thought of so much we could implement on our web app to solve the problem of food wastage. After a lot of decision making, we mainly focused on creating a useful MVC instead of trying to add so many features. This taught us lessons about focusing on the little things before targeting other ideas that though might be resourceful but won't be useful if we did not solve our major problem which for the project was to prevent food wastage by helping the user to store their food inventory and also get alerts on foods close to wastage. Things we liked as a team were the opportunities this project allowed to discover due to the fact that we had to go out of our own way to learn new things that weren't really implemented in the labs such as using API's, Render to deploy our website and also the most fun and important was the introduction to git which made using GitHub seem really resourceful we could upload code create branch and easily run our codes from each other's system easily without any complex issues though it was crazy to learn and do at first it became easier as we continued. What we dislike about this project was the little time we had to implement and create this project also though this is a common issue it reduced our productivity which was difference in availability but later we found ways to make it easier for all of us by sharing tasks we could all do at our individual tasks. What we are proud of was the fact that even after everything we built

something we are proud of and if we were to scale it would help a lot of people and even us. Also, this project brought each of us closer not just as group members but also as teammates, which made this project easier. In conclusion this project taught us that nothing is possible if you don’t start and you should always try to learn and know more without waiting to be taught or showed how to do it because that really helped us in implementing ideas and featured, we taught we never could.

THE END